Cadence: Technical Report

Contrastive Speech Analytics & Temporal Flaw Grounding · Multimodal AI Hackathon 2026, Track C

1. Problem and approach

Judges of competitive speaking (declamation, oratory, interpretive reading) score delivery inconsistently and give vague feedback. We treat delivery as a set of time series (pitch, energy, spectral balance, timing) and compare a participant against a strong reference delivery of **the same text**. Because the words are identical, forced alignment pairs every participant word with its reference counterpart. A flaw then becomes a *local, measurable deviation between two aligned time series*, which can be located to the word and explained with the numbers that caused it.

The system has four parts: (§2) a contrastive dataset with exact labels, (§3) feature extraction on forced-aligned audio, (§4) a contrastive detector that produces timed, explained flaw regions, and (§5) rubric scoring. §6 reports the stress-test evaluation, §7 describes the dashboard and §8 covers reproducibility.

2. Dataset construction

Baselines. Six excerpts (45–75 s) of landmark public-domain speeches by six speakers: FDR 1933, JFK 1961, Reagan 1981, H. R. Clinton 1995, Obama 2009, Harris 2024 (4 male, 2 female; 1933–2024 recording chains). Excerpts are continuous speech without applause; cut points are snapped to the quietest 200 ms within ±1.5 s. Audio is converted to 16 kHz mono and loudness-normalised to −23 LUFS (ITU-R BS.1770). Transcripts are drafted by Whisper and corrected by hand; word timings come from CTC forced alignment (§3.1).

Bad-mirror spectrum. Rather than hand-labelling imperfect re-recordings, we *inject* flaws into the baseline with signal processing. This makes the labels exact and the gradient controllable. Seven flaw types each have one physical parameter and a five-rung ladder from "almost perfect" (1) to "egregious" (5):

Flaw	Manipulation	Ladder (sev 1 → 5)
Monotone	TD-PSOLA; F0' = m·(F0/m)^k, m = speaker median	k = 0.70, 0.50, 0.35, 0.20, 0.05
Rushing / dragging	TD-PSOLA time-scaling, pitch preserved	×0.85 … 0.45 / ×1.2 … 2.2
Awkward pause	room-tone silence inside a phrase	0.45 … 2.2 s
Missing pause	shorten a ≥ 300 ms rhetorical pause	keep 60 % … 0 %
Volume drop	gain with 80 ms raised-cosine ramps	−4 … −19 dB
Mumbling	12th-order zero-phase low-pass, −1…−5 dB	5.0 … 1.4 kHz

Each baseline yields 35 single-flaw clips (the same word span across all five severities, a dose-response ladder) and 5 mixed clips forming an overall-quality gradient: L1 has one sev-1 flaw, and L5 has five flaws of different types at sev 4–5. Spans are cut at word boundaries (mid-gap) and spliced with 5 ms edge blending.

Exact temporal labels. Every edit returns a piecewise-linear time map (original → flawed timeline). All word times and flaw boundaries in the flawed clip are *computed through this map*, never re-estimated. Unit tests check label exactness, e.g. an inserted pause's label equals its inserted length to within 1 ms. Each clip also stores an *independent* forced alignment of the flawed audio (`aligned_words`), which is what the detector uses during evaluation. Its mean error against the exact labels is **3.8 ms** (p95 11.3 ms).

Reproducibility. All randomness (span placement, room-tone noise) is seeded by a CRC32 hash of (baseline, variant), so `build_dataset.py` regenerates the dataset byte-for-byte. Totals: 240 flawed clips with 300 labelled flaw regions (4.2 h of audio).

3. Feature extraction

3.1 Forced alignment

We use torchaudio's MMS_FA bundle, a wav2vec2 acoustic model trained for multilingual forced alignment, emitting character posteriors every 20 ms. The transcript is normalised to the model alphabet (NFKD, lower-case a–z and apostrophe), and the CTC Viterbi path gives each character a frame span; a word's start/end is its first/last character's span and its confidence is the length-weighted mean posterior. Long inputs are processed in 20 s windows with 1 s of context on each side, which bounds self-attention memory on CPU.

3.2 Frame features (10 ms hop)

Feature	Definition	Purpose
F0	Praat autocorrelation; two-pass speaker-adaptive range: floor = min(0.75·Q1, 0.6·median), ceiling = 1.5·Q3 of a 60–700 Hz first pass (after De Looze & Hirst 2008; the median cap stops the floor drifting up when part of a recording is flat); octave-jump removal (> 7 st from the 150 ms local median)	pitch contour
Pitch (normalised)	st = $12 \cdot \log_2$(F0 / median F0 of the speaker)	speaker-agnostic intonation
Energy	RMS over 25 ms Hann windows in dB, minus the speaker's median voiced level	speaker-agnostic loudness
Spectral balance	$B_2 = 10 \cdot \log_{10}(P_{2\text{–}8k} / P_{0\cdot08\text{–}2k})$, $B_4 = 10 \cdot \log_{10}(P_{4\text{–}8k} / P_{0\cdot08\text{–}4k})$ from the 512-point FFT power spectrum	consonant crispness
HNR	Praat cross-correlation harmonicity	voice clarity
Centroid, flux, MFCC-13	FFT centroid; positive log-spectral flux; CMVN MFCCs	articulation descriptors

Word-level features aggregate frames inside each aligned word: duration, syllables (orthographic vowel-group count), mean and 10–90 % range of pitch (st), power-mean level, speech-frame mean spectral balance, HNR, and the pause before the next word. Utterance summaries include speech and articulation rate, pause count/ratio, pitch s.d. and range (st), loudness range and mean HNR.

Speaker agnosticism. Pitch is in semitones relative to each speaker's own median and level is in dB relative to each speaker's own median voiced level. Every comparison (§4) then subtracts the participant's *global* offset from the reference (overall tempo ratio, pitch-spread ratio, level and balance offsets). A slower, deeper, quieter or more nasal speaker is therefore not penalised for being themselves; only local breakdowns relative to their own delivery are flagged. Global offsets are reported and scored separately (§5).

4. Contrastive detection and causal explanation

Correspondence. With identical transcripts, participant word i ↔ reference word i; otherwise words are matched by longest-common-subsequence alignment of normalised tokens. A piecewise-linear warp through matched word boundaries maps reference frames onto the participant timeline for the overlays.

Per-word deviations (signed, after removing the global offset):

- **Pace**: d = $\log_2$(Σ dur_p / Σ dur_r) over a 3-word window − global median; flagged if |d| > $\log_2(1.12)$. Each word's duration includes up to 150 ms of the silence after it: CTC places word *onsets* reliably, but where a word *ends* inside trailing silence is ambiguous (identical audio aligned as 1.12 s + 0 s gap and as 0.88 s + 0.24 s gap), and raw word durations turn that ambiguity into false tempo changes. The reported ratio uses the whole span (first onset to last offset), so compressed micro-gaps count.
- **Pitch variety**: robust spread σ = 1.4826·MAD of st over a 5-word window; d = $\log_2(\sigma_p/\sigma_r)$ − global median; flagged if 2^d < 0.78. MAD rather than s.d. makes it immune to residual octave errors.
- **Energy**: 3-word median of (L_p − L_r) − global median; flagged below −3 dB.
- **Clarity**: same construction for B_2 (tolerance 2 dB) and B_4 (4 dB); flagged when *either* band is past tolerance. B_4 catches subtle mumbling (a 5 kHz low-pass drops B_4 by ~5 dB but B_2 by only ~0.6 dB), while B_2 grades the severe end, where B_4 saturates.
- **Pauses**: expected gap = reference gap × local tempo. *Inserted*: excess > 0.3 s and gap > 0.4 s. *Missing*: reference gap ≥ 0.3 s and participant keeps < 50 %.

Temporal grounding. Flagged words form runs (bridging one-word holes; minimum 3 words for pace, 4 for pitch, 2 for energy, 3 for clarity). Edge words flagged only through window smoothing are trimmed. A region spans the first word's onset to the last word's offset, so boundaries inherit forced-alignment precision. Pause flaws are bounded by the silent gap itself (inserted) or by the two words that run together (missing). Physically coupled diagnoses are resolved, not double counted: when mumbling and a volume drop fire on the same span, the one further past its own tolerance is kept; a pause inside or touching a detected dragging (rushing) span is dropped when the span's *measured* stretch ratio accounts for the pause's length, because time-scaling also scales the silences between words.

Severity is the nearest rung of the injection ladder to the *measured* physical quantity (e.g. a measured local tempo ratio of 0.56 maps to rushing severity 4). This makes severity a statement about the signal, not an opaque score.

Causal explanation. Each region is rendered from a template into: what happened (with the quoted words and time span), an evidence table (participant vs reference vs tolerance), the formula with the actual numbers substituted, why it matters perceptually, and a concrete practice tip. For example:

> *Rushed pacing at 25.77–28.89 s* (JFK 1961, held-out; injected 25.76–28.87 s, severity 4). "in planned or accidental self-destruction We dare not tempt them with" was delivered 45 % faster than the reference (5.8 vs 3.2 syllables/s), after allowing for your overall pace. r = (Σd_participant / Σd_reference) / T_global = (3.12 / 5.65) / 1.00 = 0.55; $|\log_2 r| = 0.85 > \log_2(1.12) = 0.16$ → severity 4

Explanations are template-generated (no LLM), so reports are deterministic.

5. Scoring methodology (rubric)

Five dimensions (pace, pausing, pitch variety, vocal energy, clarity), each scored out of 100 and weighted equally:

score_d = 100 − Σ_regions P(sev)·c − G_d, clipped to [0, 100]

with P = {8, 16, 28, 42, 58} for severities 1–5. The coverage factor c = clip(T_ref/5 s, 0.6, 1.5) uses the region's *reference* duration (rushing shortens its own span). G_d is a global penalty when the whole delivery deviates beyond tolerance (tempo ±15 %, pitch spread < 75 % of the reference). Bands: ≥ 90 Excellent, ≥ 75 Strong, ≥ 60 Developing, ≥ 40 Weak, else Poor. Every deduction is itemised in the report, and the dashboard shows it per rubric bar.

6. Evaluation: stress testing across the gradient

Protocol. For each of the 240 flawed clips, the flawed audio is re-aligned from scratch, the detector is run against its baseline, and predicted regions are matched to injected ones (same type, temporal IoU ≥ 0.3, greedy one-to-one). Detector thresholds were set on two baselines (FDR, Clinton); the other four were held out.

Per flaw type and injected severity (recall; boundary errors and severity MAE over matched regions):

Flaw	Sev 1	Sev 2	Sev 3	Sev 4	Sev 5	Recall	IoU	Onset err (ms)	Offset err (ms)	Severity MAE
Monotone delivery	100%	100%	100%	100%	100%	100%	0.84	308	643	0.11
Rushed pacing	70%	100%	100%	100%	100%	93%	0.90	107	189	0.05
Dragging pacing	100%	100%	100%	100%	100%	100%	0.94	178	201	0.05
Awkward mid-phrase pause	100%	100%	100%	92%	100%	98%	0.92	34	43	0.00
Missing rhetorical pause	0%	89%	100%	100%	100%	80%	0.98	6	8	0.17
Volume drop	100%	100%	100%	100%	100%	100%	0.98	88	5	0.04
Mumbled articulation	33%	100%	100%	100%	100%	90%	0.98	3	61	0.42

Overall: recall **94 %**, mean temporal IoU **0.93**, mean onset / offset error 110 / 177 ms, **0.14 false positives per minute of audio**, severity MAE ≤ 0.17 for every type except mumbling (0.42). Overall score vs gradient level L1–L5: Spearman ρ = **−0.96** (single-flaw ladders −0.88 to −0.95).

Per speech: tuning vs held-out:

Baseline	Set	Recall	IoU	Onset err (ms)	Offset err (ms)	False pos./min
`clinton_1995`	tuning	94%	0.94	127	98	0.05
`fdr_1933`	tuning	92%	0.95	71	104	0.23
`harris_2024`	held-out	96%	0.93	146	290	0.25
`jfk_1961`	held-out	92%	0.96	53	189	0.08
`obama_2009`	held-out	94%	0.90	172	163	0.22
`reagan_1981`	held-out	98%	0.93	92	214	0.07

Held-out speeches only: recall 95 %, IoU 0.93, 0.15 false positives/min, the same as the tuning set.

Development history (for transparency). The first full run with frozen thresholds gave recall 94 %, IoU 0.93 and **0.25 false positives/min** (held-out 0.28/min; archived in `docs/results/evaluation_before_silence_fix.*`). Inspecting the held-out false positives exposed three systematic measurement errors, which we fixed. None of the fixes changed a threshold: (1) word-end ambiguity inflating tempo deviations (16 of 20 spurious "rushing" flags sat on one JFK

phrase), fixed by the 150 ms silence allowance in §4; (2) pauses stretched by a detected dragging span reported a second time, fixed by the coupling rule in §4; (3) the adaptive pitch floor rising when a long span was flattened, deleting genuine low notes elsewhere (a recurring Harris "monotone" flag), fixed by the median-capped floor in §3.2. Because these fixes were found on the held-out speeches, the post-fix held-out numbers are not strictly blind; the pre-fix run is the blind result.

Reading the results. Detection is near-complete from severity 2 upward for every flaw type, and severity is recovered almost exactly because the detector measures the same physical quantity the injector controls (a rushed span measured at ratio 0.55 against an injected 0.55). The misses cluster at the subtlest rungs, by design: a *missing pause* at severity 1 keeps 60 % of the pause, above the 50 % tolerance a listener would also accept, and *mumbling* with a 5 kHz low-pass is physically invisible on recordings with little energy above 4 kHz (FDR 1933). Pause regions are the most precise (≈ 5–45 ms boundary error) because they are bounded by aligned silence; monotone regions are the loosest (onset 0.3 s, offset 0.6 s) because pitch spread is measured over a 5-word window, which spreads the evidence into neighbouring words. An identical delivery produces no flaws (unit test), and forced alignment of the flawed audio stays within 3.8 ms of the exact labels on average, so alignment is not the limiting factor.

Cross-speaker test (self-recordings): a negative result. Two team members each read the Harris excerpt three times on consumer devices: a natural `good` take and two takes performing seven directed flaws at marked word spans (§2; labels from forced alignment of each recording). With the thresholds above, the detector found **9 of 14** directed flaws (all pauses and missing pauses, both monotone and rushing spans, 1 of 2 dragging spans; neither mumbling nor volume drop), but it also flagged **about 30 regions per take, including the natural readings**: 57 false positives per minute, with natural takes scoring 31/100, no better than the flawed takes. The cause is measurable. In synthetic clips every word outside the injected flaw is identical to the reference (deviation 0.0× tolerance), whereas between two real readers the *median* word already deviates 1.4–3.4× tolerance in pace, pitch spread and spectral balance, and the readers place pauses differently (10–15 vs 21 pauses ≥ 0.3 s). The synthetic benchmark therefore measures *localisation of a flaw given an otherwise identical delivery*; it does not measure robustness to natural reader-to-reader variation, and the word-by-word tolerances are far too tight for it.

Attempted fix: speaker-adaptive tolerances (not adopted). We widened every tolerance to 3 × the robust spread (1.4826 · MAD) of the recording's own per-word deviations, so only spans that stand out from the reader's *own* variability are flagged; on synthetic clips that spread is ≈ 0 and the benchmark was unchanged (recall 94.3 %, IoU 0.93). On the self-recordings it cut false positives by 75 % (57 → 14.5 per minute) and natural takes rose to 74/100, but directed flaws found fell from 9/14 to 3/14. Sweeping the multiplier from 1 to 3 (chosen on one speaker, checked on the other) gave no acceptable trade-off: performed flaws are about as large as natural style differences. We therefore kept the original detector; the experiment is preserved on the `speaker-relative` branch.

7. Dashboard

React + TypeScript (Vite), wavesurfer.js and ECharts, served by a FastAPI backend.

- **Contrastive dataset explorer**: pick a speech, then any rung of the overall gradient (L1–L5) or any cell of the 7 × 5 single-flaw matrix. The *injected vs detected* strip shows ground truth and predictions on one timeline.
- **Analyse your delivery**: choose a reference (dataset speech or your own upload plus transcript), then record in the browser or upload any audio/video file (decoded by ffmpeg).
- **Results**: overall score with itemised rubric bars; waveform with severity-shaded flaw regions (click to replay); three small-multiple overlays (pitch st, loudness dB, spectral balance dB) of participant vs time-warped reference with a linked crosshair, zoom, and playhead; flaw cards with *Play yours* / *Play reference* A/B audio, the evidence table and the formula; a forced-aligned transcript that follows playback.
- **Stress test** page: the §6 recall heat-map and summary tiles.

8. Reproducibility and engineering

- Exact Python dependencies in `uv.lock` (CPU-only torch on all platforms); Node dependencies in `package-lock.json`; a single Dockerfile builds the dashboard and serves both on port 8000.
- Deterministic dataset build (seeded), deterministic analysis (no sampling; templates instead of generation); unit tests for label exactness and end-to-end detection on synthetic audio.
- The full dataset is rebuildable from the original public-domain sources with three commands.

9. Limitations and next steps

Injected flaws are perfectly controlled but simplified: real rushing also slurs consonants, and real nerves add pitch tremor. The cross-speaker test (§6) shows the main open problem: word-by-word deviations from a single reference speaker confound *style* with *flaws*. Two fixes follow directly: (1) judge local delivery against the participant's **own** global behaviour (a span is rushed if it is much faster than *this speaker's* median rate) and use the reference only for global comparisons, and (2) calibrate tolerances on natural readings by many speakers instead of synthetic copies. Thresholds are currently literature- and data-calibrated constants, not learned. With a larger set of human recordings, a learned per-dimension classifier on the same word-level deviation features would be a natural extension. Fillers ("um", "uh") and mispronunciations are not yet modelled. A transcript mismatch is handled by sequence alignment, but unmatched words are excluded from comparison.

References

- De Looze, C., & Hirst, D. (2008). Detecting changes in key and range for the automatic modelling and coding of intonation. *Speech Prosody*.
- Moulines, E., & Charpentier, F. (1990). Pitch-synchronous waveform processing techniques for text-to-speech synthesis using diphones. *Speech Communication*, 9(5–6).
- Pratap, V., et al. (2023). Scaling speech technology to 1,000+ languages (MMS). *arXiv:2305.13516*.
- Boersma, P. (1993). Accurate short-term analysis of the fundamental frequency and the harmonics-to-noise ratio of a sampled sound. *IFA Proceedings 17*.
- ITU-R BS.1770-4 (2015). Algorithms to measure audio programme loudness and true-peak audio level.
- Jadoul, Y., Thompson, B., & de Boer, B. (2018). Introducing Parselmouth: A Python interface to Praat. *Journal of Phonetics*, 71.